

The Planner Is the Adversary: Test-Time Search Exploits Learned Costs in Latent World-Model Planning

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Abstract

Action-conditioned JEPA world models plan by test-time search: at every control step, a sampling optimizer (CEM) scores hundreds of imagined futures with a learned cost and executes the best candidate. We show that on contact-rich manipulation this coupling — an optimizer pitted against a learned proxy — is where planning fails, and that the dominant and previously unmeasured failure mode is *cost overoptimization* (Goodhart’s law) rather than merely missing information. All experiments use frozen pretrained checkpoints. Two kinds of numbers recur: *planning success*, counted as successful episodes out of 16 (written $k/16$), and *readout accuracy*, the percentage of frames whose decoded object position falls within the 5 cm task-success radius. **(1) Diagnosis.** We measure action grounding with counterfactual ranking accuracy (CRA): given a real transition, does the predictor place the executed action’s imagined future closer to the observed next state than the futures it imagines for 16 distractor actions? Chance is $1/17 \approx 6\%$. Against distractors drawn from similar states, every model sits at the chance floor in contact regimes — and the result is flat across a $45\times$ encoder scale-up (DINOv2 22M $\rightarrow$ V-JEPA-2-AC 1B). **(2) Localization.** An *oracle ladder* fixes the planner and its budget, gives it *perfect dynamics* (every candidate action sequence is rolled in the simulator and encoded by the real frozen encoder), and varies only the cost. With the true simulator-state cost, the same planner solves push 16/16; with every cost computed from the frozen representation it solves at most 2/16. Even repairing the object readout on exactly the frames the planner scores (readout accuracy 78% $\rightarrow$ 91.5%) adds *zero* successful episodes. **(3) Mechanism.** The planner is the adversary: on the elite candidates CEM converges to, the same readout that is 91.5% accurate on random frames drops to 24% — the search seeks out precisely the frames the cost mis-scores, and executed plans read “at goal” while the true object sits 13–30 cm away. Sweeping the search budget makes this monotone in optimization pressure: proxy improving, true outcome flat, readout error on the elites growing — a Goodhart curve with an honest-cost control that, in contrast, converts search into success. Mitigations from the reward-hacking playbook — readout hardening, encoder-LoRA fine-tuning (5 seeds), ensemble disagreement penalties — all stay at $\leq 2/16$, even while readout accuracy reaches 94–100%: *grounding does not buy planning*. Overoptimization curves confirm the mechanism is monotone in search budget, and a search-free amortized controller — removing the optimizer entirely — also fails (0/8), exposing a second, independent wall: the frozen representation. Grounding is necessary, not sufficient. **(4) The predictor-side lever.** An action-counterfactual InfoNCE objective (a 0.5M-parameter predictor-LoRA, no object labels) lifts CRA on real-robot DROID data from the 6% chance floor to 61% and cuts a planning-probe action error by 26%, while *improving* reconstruction. The implication for evaluation: prediction accuracy and grounding metrics do not predict plannability; world models intended for planning should report the cost’s *exploitability under the planner’s own optimization pressure*.

1 Introduction

Action-conditioned world models that predict in a learned latent space — the JEPA family (DINO-WM (Zhou et al., 2024), V-JEPA-2-AC (Assran et al., 2025), and the design study of (jepa-success authors) (2025)) — plan by coupling two learned components with a test-time optimizer: a predictor imagines futures, a cost scores them against a goal, and a sampling planner (CEM) searches for the action sequence that minimizes the cost. On free-space tasks this works well. But a consistent pattern runs through the field’s own evaluations:

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the *positive* closed-loop numbers are for reaching and pose control, while contact-rich tasks — pushing, grasping, pick-and-place — are omitted, demoted to qualitative figures, or propped up with hand-crafted sub-goals. (jepa-success authors) (2025) stop their closed-loop Metaworld tables at Reach and note the model “hallucinates grasping the object”; Assran et al. (2025) report pick-and-place only after supplying three manual sub-goal images.

The field’s implicit diagnosis is a *model-capacity* story: the encoder is too small, the representation too coarse, the predictor too weak — scale will fix it. This paper argues, and then shows with a chain of controlled experiments, that the binding failure is somewhere else entirely: **in the optimizer–cost interface**. A CEM planner at test time is an optimizer pitted against a learned proxy objective. The reinforcement-learning-from-human-feedback literature has a name for what happens to optimizers pitted against learned proxies: *overoptimization*, or Goodhart’s law — push on a proxy hard enough and the true objective decouples from it (Gao et al., 2023; Skalse et al., 2022; Karwowski et al., 2024; Amodei et al., 2016). We show the same phenomenon governs latent world-model planning, with one aggravating difference: in RLHF the optimizer is a training loop one can regularize; here the optimizer runs *at every control step*, on frames the cost was never fit on, selected adversarially *because* the cost mis-scores them.

Three features distinguish our account from the classic “model exploitation” concern in model-based RL (Janner et al., 2019; Lambert et al., 2020), where a policy exploits errors in learned *dynamics*. (i) We give the planner *perfect dynamics* — every candidate action sequence is rolled in the simulator and re-encoded by the real frozen encoder — and the failure persists unchanged; the exploited component is the *cost*, cleanly isolated. (ii) We show the exploitation is *irreducible* for the frozen-encoder regime: a ladder of increasingly strong cost repairs (better formula, better readout, off-policy hardening, encoder fine-tuning, ensembling) all fail the same way. (iii) We show the standard diagnostic quantities the field might use to catch this — prediction error, action-grounding scores, readout precision — are all satisfied while planning fails, i.e. the failure is invisible to current evaluation practice.

Concretely, our experimental chain (Metaworld with simulator ground truth for supervision and verification; DROID for real-robot transfer) is:

- **C1 (diagnosis at scale, Sec. 3).** On frozen checkpoints, action grounding against *nearest-neighbour* action distractors sits at the chance floor (≈ 0.059) in every contact-relevant regime, and our Boundary-Blindness instrument shows the models smear exactly the contact bifurcations that decide a grasp. Crucially this is *scale-invariant*: flat across a $45\times$ encoder scale-up and two self-supervised families, up to and including V-JEPA-2-AC at 1B parameters.
- **C2 (the wall, at upstream parity, Sec. 4).** Our closed-loop harness reproduces the published Metaworld Reach baseline ($6/16 = 37.5\%$, CI $[18.5, 61.4]\%$, vs. the reported $44.8 \pm 8.9\%$), and both the L_2 and grounded planners score $0/16$ on push and pick-and-place with a diagnostic signature: the arm arrives (2–4 cm from its goal pose), the object never moves.
- **C3 (localization: the oracle ladder, Sec. 5).** Swapping one component per rung under perfect dynamics: the true-state cost solves push $16/16$; latent- L_2 , object-readout, learned-metric, and probe-based state costs collapse to $0\text{--}2/16$. A measured repair of the off-policy readout ($78 \rightarrow 91.5\%$ within 5 cm on the exact frame distribution CEM scores) transfers *zero* planning success — readout precision was never the binding constraint.
- **C4 (mechanism: the planner is the adversary, Sec. 6).** CEM’s converged elite population decodes at 24% within 5 cm versus 91.5% on random off-policy frames — the search specifically finds and converges onto the cost’s error pockets, driving the object to where the cost reads “at goal” while it is 13–30 cm away. Sweeping the search budget makes this a Goodhart curve monotone in optimization pressure — proxy improving, true outcome flat, elite readout error growing — with an honest-cost control (reach) that instead turns search into success ($2/8 \rightarrow 8/8$). Mitigations from the reward-hacking playbook — hardened readouts, encoder-LoRA (5 seeds), ensemble disagreement penalties, relearned representation adapters — all stay within the frozen baseline’s $0\text{--}2/16$, even at 94–100% grounding: *grounding does not buy planning*.
- **C5 (removing the adversary is not sufficient, Sec. 8).** A search-free amortized controller (goal-conditioned inverse dynamics on the same frozen representation) also solves push $0/8$, and re-adding

search in doses corrupts its plans at a rate rising with dose ($0 \rightarrow 7\%$) but never crosses the wall — the object never moves. Contact planning faces *two* walls: the exploitable cost/optimizer interface and an insufficient frozen representation. Grounding is necessary, not sufficient.

- **C6 (the axis that moves, Sec. 7).** An action-counterfactual InfoNCE objective on the predictor (LoRA, 0.5M parameters, no object labels) improves DROID Action-Error in all six regime $\times$ horizon cells (pooled $1.468 \rightarrow 1.086$, -26%) and lifts effect-conditioned action grounding from chance (0.061) to 0.605, while improving reconstruction — the training-side counterpart of the analysis, and the correct level at which to intervene on the *predictor* axis.

Conventions (how to read our numbers). Four kinds of quantities appear throughout, and we keep their formats distinct. *Planning success* is always episodes solved out of 16 paired evaluation episodes, written $k/16$ (with the percentage in parentheses when comparing to published baselines). *Readout accuracy* is always the percentage of frames whose decoded object position falls within the 5 cm task-success radius (“% within 5 cm”), with the median error in cm where informative. *CRA* is a ranking accuracy over transitions, reported as a fraction in tables; with 16 distractors, chance is $1/17 \approx 0.059$. *AUG/ECS* are raw latent-space gaps (Sec. 3.1), meaningful only within a model. Distances in the workspace are in cm or in Metaworld state units where noted.

The through-line is a reframing of what world-model evaluation must report. Prediction error is satisfied while planning fails (Sec. 3); grounding is satisfied while planning fails (Sec. 6); readout precision is satisfied while planning fails (Sec. 5). The quantity that tracks the failure is the gap between what the cost claims and what the world delivers *on the states the optimizer converges to* — the cost’s exploitability under optimization pressure. We propose it as a first-class evaluation axis for world models intended for planning.

2 Related work

JEPA world models for planning. DINO-WM (Zhou et al., 2024) predicts forward in the frozen feature space of a self-supervised image encoder and plans with sampling-based MPC from a goal image; V-JEPA-2-AC (Assran et al., 2025) learns an action-conditioned world model on a video encoder and demonstrates zero-shot manipulation; (jepa-success authors) (2025) systematically study the design choices that make such models plan. We take these as frozen baselines and treat their own evaluations as evidence: across all three, strong closed-loop results are pose/free-space tasks, and contact is where the story thins (hand-crafted sub-goals, qualitative figures, “hallucinated grasping”). Our work explains why the wall sits precisely at contact and why it does not yield to scale.

Overoptimization and reward hacking. Optimizing against a learned proxy eventually decouples proxy from truth: formalized for RLHF reward models by Gao et al. (2023), characterized in general by Skalse et al. (2022) and Karwowski et al. (2024), and anticipated as “reward hacking” (Amodei et al., 2016). We port this lens to latent-space planning, where it has not been quantitatively established: the CEM inner loop is the optimizer, the latent cost is the proxy, and the elite population is the analogue of the over-optimized policy. Our oracle ladder gives this port an isolation the RLHF setting lacks — we can hold dynamics *perfect* and vary only the cost.

Model exploitation in model-based RL. That policy optimization exploits errors in learned *dynamics* is a classic MBRL concern (Janner et al., 2019), related to the objective mismatch between likelihood training and control performance (Lambert et al., 2020). Our finding is materially different: with dynamics oracle-controlled, the *cost* is exploited at test time — a failure that persists when the dynamics-error channel is closed entirely, and that standard dynamics-side remedies (ensembles, pessimism) do not touch (Sec. 6).

Concurrent diagnostics and objectives. Concurrently, ATM ((ATM authors), 2026) proposes a post-hoc action-identifiability diagnostic for frozen latent world models with a training-time supervision fix, and UWM-JEPA ((UWM-JEPA authors), 2026) shows in a physics toy setting that teacher-forced JEPA objectives admit action-invariant solutions which counterfactual targets repair — independently confirming

the mechanism behind our Sec. 7 objective. Neither work addresses planning: our contributions beyond both are the planner-level exploitation analysis (Secs. 5–6), the scaling study, and the demonstration on pretrained robot world models at up to 1B parameters. IMWM ((IMWM authors), 2026) observes that finite-budget sampling planners can be the bottleneck even under an idealized world model and responds with demonstration priors; (authors) (2026b) amortize planning into an inverse-dynamics controller, removing test-time search — a direction our analysis independently motivates as *removing the adversary* (Sec. 8). Vision-tactile world models ((authors), 2026a) address what information contact planning needs; our results say that adding information without addressing exploitability is not sufficient (Sec. 5).

Benchmarks. We evaluate on Metaworld (Yu et al., 2020), whose simulator state lets us supervise and *verify* costs against ground truth, and DROID (Khazatsky et al., 2024) as the real-robot transfer check. Planning follows the upstream CEM configuration (Rubinstein, 1997).

3 Diagnosis: action grounding is at chance and does not scale

We first establish, on frozen checkpoints (nothing trained), that the contact-regime action grounding the planner would need is absent — and that this is not a capacity artifact. This section compresses the diagnostic instrument; full definitions and the AUG/ECS corroboration are in App. A and App. D.

3.1 Metrics

Let $F(z, a)$ be the frozen one-step latent predictor (driven through the model’s own `unroll` primitive, with the model’s own action normalisation) and d the upstream planning distance (L_2 for every baseline, matching the shipped `L2_cem` configuration). We measure three complementary quantities on held-out factual transitions (z_t, a_t, z_{t+1}) ; each answers a different question.

CRA — does the model rank the executed action best? Given the transition, we sample $K=16$ distractor actions $\{a_k^-\}$ and ask whether the factual action’s prediction lands closer to the realised next latent than every distractor’s prediction:

$$\text{CRA} = \Pr \left[d(F(z_t, a_t), z_{t+1}) < \min_{k=1\dots K} d(F(z_t, a_k^-), z_{t+1}) \right], \quad (1)$$

so chance is $1/(K+1) = 1/17 \approx 0.059$. The distractor family controls what the metric tests: *opposite* distractors ($a^- = -a_t$) test coarse directional grounding — the quantitative version of the “open vs. close the gripper” visualisations in prior work — while *nearest-neighbour* distractors are actions *actually executed* from the K most similar latent states in the dataset: plausible alternatives a competent policy might take from this state, so ranking the factual action above them requires resolving exactly the fine action differences planning depends on. Being a ranking, CRA is comparable across models with different latent scales; it is our primary cross-model metric.

AUG — does the action help at all? The Action Usage Gap is the prediction-error increase when the action is replaced by another action from the batch (a permutation π):

$$\text{AUG} = \mathbb{E} \left[d(F(z_t, \pi(a_t)), z_{t+1}) - d(F(z_t, a_t), z_{t+1}) \right], \quad (2)$$

which is ≈ 0 if and only if the predictor ignores its action input. AUG is a raw latent-space gap: informative *within* a model, not comparable *across* models (latent scales differ), which is why it corroborates CRA rather than replacing it (App. D).

ECS — the same question, asked where it matters. Averaged over all transitions, AUG can be diluted by near-static steps (the arm hovering; the scene unchanged) where different actions genuinely lead to similar futures. Effect-Conditional Sensitivity restricts AUG — and, in the *effect-conditioned CRA* we report as the primary signal, the ranking — to transitions that actually move the latent, $\|z_{t+1} - z_t\| > \tau$ with τ the per-model median displacement. A low effect-conditioned score therefore reflects genuine action-ignoring, not quiet data.

Boundary Blindness (BB; definition in App. A) is our fourth instrument, answering a question none of the above can: is the model’s action-sensitivity *calibrated to the world’s*? On a neighbourhood of similar

states, it compares how much the *world's* true outcomes fan out under nearby actions against how much the *model's* predictions fan out under the same actions, both z-scored over the population; $BB > 0$ means the world bifurcates (e.g. grasp vs. miss) while the model predicts nearly the same future for every nearby action. Because BB references ground-truth outcomes, a high value cannot be explained away by small one-step latent deltas.

Protocol. Transitions are stratified into four regimes — free space, pre-grasp, gripper actuation, contact manipulation — via a proxy from the Metaworld simulator state (object displacement as the contact signal; DROID uses proprioception and latent-change heuristics). All confidence intervals are trajectory-clustered bootstrap. Every frame is encoded once and cached; nothing is trained for the diagnostic.

3.2 Frozen baselines fail on nearby actions, at the boundary

Table 1: Effect-conditioned CRA (top-1, n_{eff} -weighted pool, Metaworld). Against *opposite* distractors the models are near-perfect — they use actions. Against *nearest-neighbour* distractors they collapse, worst at pre-grasp: they cannot resolve the *small* action differences that decide a grasp.

model	distractor	free space	pre-grasp	contact
DINO-WM	opposite	0.990	0.961	0.975
	nearest-neighbour	0.566	0.467	0.481
JEPA-WM	opposite	0.993	0.972	0.984
	nearest-neighbour	0.635	0.541	0.571

Boundary Blindness explains and localizes the collapse: pooled BB_{boundary} concentrates at the pre-grasp boundary on both model families (DINO-WM 1.323, JEPA-WM 1.280, vs. free-space 0.28–0.30; per-task CI-aware pairing elevated in 4/6 and 5/6 tasks with zero confident reversals) and transfers to real-robot DROID data (1.975 [1.601, 2.350] under the $\|\Delta z\|$ proxy). The predictor is a conditional-mean estimator under full-latent L_2 ; where contact bifurcates the outcome fan, it predicts the fan's mean.

3.3 The failure does not scale: 45 $\times$, two families

We score four frozen DROID world models on an identical 333-episode subset: DINO-WM (DINOv2 ViT-S, 22M), JEPA-WM (DINOv3 ViT-L, 300M), V-JEPA-2-AC (ViT-G, 1.01B), and the independent public V-JEPA-2-AC-OSS (1B).

Table 2: DROID action grounding does not scale. *Top*: effect-conditioned CRA against nearest-neighbour distractors (chance = $1/17 \approx 0.059$): all four models sit at the chance floor in every action-critical regime. *Bottom*: $BB_{\text{boundary}} > 0$ in every cell (boundary-blind), pre-grasp locus in 3/4. A positive control rules out metric saturation: on toy fully-actuated datasets the same diagnostic returns 0.66–1.0.

regime	DINO-WM 22M	JEPA-WM 300M	V-JEPA2-AC 1B	V-JEPA2-AC-OSS 1B
<i>effect-CRA, nearest-neighbour (chance ≈ 0.059)</i>				
pre-grasp	0.068	0.077	0.061	0.069
gripper actuation	0.031	0.059	0.049	0.031
contact manipulation	0.055	0.045	0.035	0.033
$BB_{\text{boundary}} (> 0 = \text{boundary-blind})$				
pre-grasp	1.916	1.205	1.933	1.560
free space	0.963	1.103	1.010	0.819
gripper actuation	1.088	1.057	0.643	0.828
contact manipulation	0.421	0.897	0.390	0.595

The curve is flat across 45 $\times$ scale and across families. “Just scale the encoder” is ruled out; whatever fixes contact planning, it will not be free with the next checkpoint. Note what this section does *not* yet

establish: that this grounding failure is what breaks planning. Sections 5 and 6 show the relationship is more damning — even *repairing* grounding does not repair planning.

4 The wall, at upstream parity

We match the upstream Metaworld closed-loop evaluation read off its shipped configuration: goal = the scripted expert’s final frame; CEM with horizon 6, 300 samples, 15 iterations, executing 3 model-steps per replan, ≤ 100 env steps; success = the simulator’s flag at episode end. Reaching a faithful protocol required fixing three environment-side reproduction bugs, documented with verification probes in App. C (after the fixes, rendered observations are on the training manifold: one-step prediction error $1.4\times$ dataset level, latent NN ratio 0.97; physics identical, 1.5 mm replay error).

Table 3: Closed-loop Metaworld, 16 paired episodes per task (paired env initialisation and CEM noise; episode-end judging). The harness reproduces the published Reach baseline; contact collapses with the diagnostic signature.

task	arm	success	final state-dist	final ee-dist
mw-reach	L_2	6/16 (37.5%)	0.332	0.022
	grounded	8/16 (50.0%)	0.412	0.031
mw-push	L_2	0/16	0.624	0.038
	grounded	0/16	0.527	0.028
mw-pick-place	L_2	0/16	0.578	0.035
	grounded	0/16	0.496	0.021

Three readings. (i) Where the diagnostic is clean (free space), the harness reproduces the published baseline (Reach 6/16, i.e. 37.5% with CI [18.5, 61.4]%, vs. the reported $44.8 \pm 8.9\%$) — the harness is healthy, the contact failure is real. (ii) The failure signature is specific: final end-effector 2–4 cm from its goal pose while the object never moves — the closed-loop face of the smeared contact bifurcation, matching the upstream reports (“hallucinates grasping”; sub-goal images required). (iii) A grounded auxiliary cost (a 0.5M object-dynamics channel $h(z, a) \rightarrow \Delta \text{obj}$ whose construction and metric-level gains we defer to App. B) improves the cost surface with a CI-clean paired gain (final state-dist +0.089 [+0.022, +0.162] pooled contact) — *and flips zero successes*. That last observation is the loose thread this paper pulls: if the model is the problem, why does repairing the model’s outputs not repair planning?

5 The oracle ladder: the wall is the cost

We isolate which component breaks contact planning by swapping exactly one rung at a time, holding everything else at the upstream protocol (CEM 100×6 , $H=6$, strict episode-end success, 16 episodes, `dino_wm_metaworld`). The key device is the *latent oracle*: every CEM candidate is rolled in the *simulator*, rendered, and encoded by the real frozen encoder — perfect dynamics through the true representation, learned predictor removed. Any rung that fails under the latent oracle fails because of its *cost*, full stop.

Read from the bottom up, the ladder rules out, in order: the **predictor** (perfect dynamics changes nothing: 0/16 either way); the **planner and budget** (the same CEM solves 16/16 given the true-state cost); the **cost formula** (the probe-based rung copies the successful cost’s exact functional form, including the hand-approach shaping term, and still collapses $16 \rightarrow 2$); and **static readout precision** (a spatial probe decodes the object to 92% within 5 cm on held-out frames — the encoder *carries* the object position within the success radius).

The decisive rung: a real readout repair transfers nothing. One hypothesis survives the ladder: CEM scores *off-policy* frames (latents of arbitrary candidate rollouts), where an expert-trained probe was never fit. This is true — the expert probe degrades to 78% within 5 cm on random off-policy frames — and it is fixable: retraining the probes with off-policy frames mixed in restores **91.5%** within 5 cm (push

Table 4: The oracle ladder (Metaworld push / pick-and-place, strict episode-end success, 16 episodes per rung except the object-readout and learned-metric rungs, run at 8). Same planner, same budget, same success radii on every rung; each rung changes one component. Perfect dynamics + the true-state cost solves the task; perfect dynamics + *any* frozen-encoder cost does not.

rung	dynamics	cost	readout	push	pick
state oracle	perfect	true state ($\ obj-g\ + 0.5\ hand-obj\ $)	sim	16/16	11/16
latent oracle	perfect	latent L_2	—	0/16	0/16
latent oracle	perfect	object readout ($\ g(z)-g(z_g)\ $)	probe	0/8	0/8
latent oracle	perfect	learned metric $d_\theta(z, z_g)$	—	0/8	0/8
latent oracle	perfect	<i>true-state cost form</i> , probe readouts	probe	2/16	0/16
latent oracle	perfect	same, <i>off-policy-hardened</i> probes	probe	1/16	0/16
world model	learned F	latent L_2 (upstream)	—	0/16	0/16

92.9%, median 2.0 cm) *on the same frame distribution the planner scores*. The re-gate: push 1/16. A measured, targeted repair of the hypothesized binding constraint — readout precision on the planner’s own query distribution — transferred *zero* planning success. Off-policy readout precision was never the binding constraint. Something about the *interaction between the optimizer and the cost* is.

6 The planner is the adversary

6.1 Direct evidence: CEM converges onto the cost’s error pockets

The reconciliation of Sec. 5’s paradox is that CEM does not sample off-policy frames *uniformly* — it searches for the cost *minimum* over 100×6 candidates per replan, so any residual-error pocket in the cost is found and converged upon, precisely *because* it scores too low. We test this by decoding the probe on the CEM *elite* population itself — the frames the planner actually trusts — mined from the same hardened-probe runs:

Table 5: Object decode error of the *off-policy-hardened* probe on the CEM elite distribution (13,440 elites; final iteration = the converged population), vs. the same probe on random off-policy frames. The probe is accurate where CEM does not look and inaccurate exactly where it converges.

frame distribution	median err	within 5 cm	within 7 cm
random off-policy (reference)	2.0 cm	91.5%	—
all CEM elites	7.0 cm	24.0%	50.5%
final-iteration elites	7.3 cm	19.2%	46.0%
final-iteration, push	7.7 cm	15.3%	38.9%

The same probe that reads 2 cm on random frames reads 7–8 cm — beyond the 5 cm success radius — on the frames its own planner selected. The closed-loop consequence is visible in the executed episodes: the cost reports near its minimum (“object at goal”) while the true object sits 13–30 cm away. This is reward hacking in the strict sense (Skalse et al., 2022): the optimizer achieves a low proxy value by exploiting the proxy’s misspecification, not by achieving the goal.

6.2 Overoptimization curves

The framing makes a falsifiable prediction the previous sections do not test: exploitation should be *monotone in optimization pressure*. Holding everything fixed and sweeping the CEM budget, an honest cost should convert extra search into better true outcomes; an exploitable cost should convert it into deeper pockets — proxy improving, truth flat or worse, in direct analogy to the reward-model overoptimization curves of Gao et al. (2023). We test this on the latent oracle (perfect dynamics; only the cost is learned, so any proxy/truth

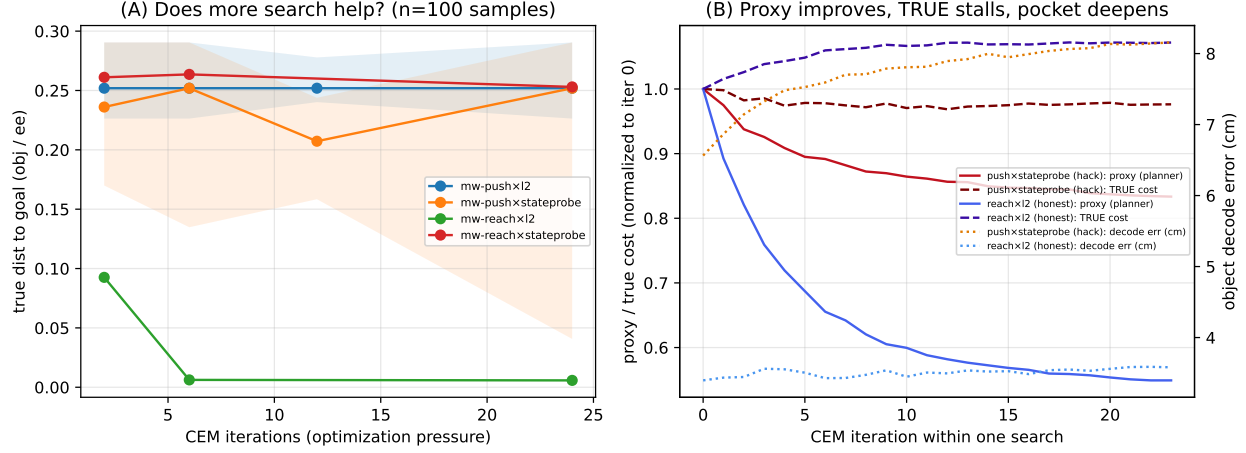


Figure 1: **Cost overoptimization in latent planning.** (A) True distance to goal vs. CEM iterations. reach $\times L_2$ (honest cost) converts more search into success (2/8 $\rightarrow$ 8/8); push $\times$ probe-cost (exploitable) and push $\times L_2$ (no-signal) do not improve with budget. (B) Within one search at the largest budget, normalized to the first CEM iteration: for push $\times$ probe-cost the proxy the planner minimizes falls (-17%) while the true state cost stalls (-2%) and the object-decode error on the committed elites *grows* (6.6 $\rightarrow$ 8.2 cm, $+24\%$) — the planner digs into deeper readout-error pockets. The honest cost (reach $\times L_2$) shows no such decode growth.

divergence is attributable to the cost under optimization, not to predictor error), sweeping CEM iterations $\in \{2, 6, 12, 24\}$ over $n=8$ paired episodes, with three cost $\times$ task arms that isolate the three regimes.

The three predictions all hold (Fig. 1). **Exploitable cost** (push, probe-based state cost): within a single search at budget 24, the believed-best candidate’s proxy falls monotonically (0.166 $\rightarrow$ 0.138, -17%) while its true object $\rightarrow$ goal distance is flat (0.279 $\rightarrow$ 0.273, -2%) and the probe’s decode error on the elite population grows monotonically (6.6 $\rightarrow$ 8.2 cm, $+24\%$): more optimization, deeper pockets. Across budgets the true object distance does not improve with search (med $\approx$ 0.25 m at all of $\{2, 6, 12, 24\}$ iterations; episode success stays within the frozen band with CIs overlapping 0). **Honest cost** (reach, L_2): the essential positive control — here more search *does* convert to success (2/8 $\rightarrow$ 8/8 $\rightarrow$ 8/8 at 2/6/24 iterations), confirming the sweep detects “search helps” when the cost is aligned. **No-signal cost** (push, L_2): the proxy falls (-32%) but the true object distance is *exactly* flat (0.252 m at every budget) — L_2 -in-latent has no minimum at task success, so optimizing it is inert; this distinguishes an *exploitable* cost (signal present, worst-case reachable) from a merely *uninformative* one. A second pressure axis — CEM population $\in \{50, 100, 300\}$ at fixed 12 iterations — corroborates: enlarging the candidate pool does not improve the true push outcome either (probe-cost median 0.233/0.207/0.252 m; L_2 flat).

6.3 Mitigations from the playbook all fail

If the diagnosis is right, repairs that improve the cost’s *average* quality without removing its *worst-case* pockets should not cross the wall — the optimizer finds whatever pockets remain. We ran the standard playbook, each arm under the latent oracle (perfect dynamics), push held-out, strict success:

Three of these deserve comment. **Encoder fine-tuning** (LoRA on the DINOv2 trunk, action-conditioned grounding objective, 5 seeds): grounding reaches 94–100%, the five-seed planning mean is 1.8/16 with CI $[0, 3.7]$ — statistically inside the frozen band; the single 5/16 seed is high-variance noise, not a crossing. This is the strongest form of the paper’s recurring lesson: *grounding does not buy planning*, now proven at the encoder level, not just the readout level. **Ensemble disagreement** (cost = $\text{mean}_k d_k^2 + \lambda \text{Var}_k [d_k]$ over 5 independently fine-tuned seeds) is the standard pessimism remedy for model exploitation (Janner et al., 2019); it fails here ($\{2, 2, 2, 1\}/16$ across λ) for an instructive reason: the seeds share the frozen pre-trained base, so they share the blind spot, and disagreement stays flat exactly on the exploited pockets — the same failure mode reported for reward-model ensembles sharing a pre-trained trunk. **The state-oracle row** pins

Table 6: Mitigation ladder (latent oracle, push; 16 episodes per arm except the learned-metric arm at 8). Every intervention that operates on or above a frozen-base representation stays inside the frozen baseline’s 0–2/16 band — including those that push readout accuracy to near-ceiling. The true-state cost row shows the same planner and budget solve the task when the cost has no pockets.

intervention	grounding-level metric	push
latent L_2 (frozen null)	—	0/16
probe-based state cost	static decode 92% <5 cm	2/16
off-policy-hardened probes	off-policy 91.5% <5 cm	1/16
learned latent metric d_θ	—	0/8
relearned repr. adapter φ (v2)	held-out decode med. 6.9 cm	1/16
encoder-LoRA + φ (5 seeds)	decode 94–100% <5 cm	{5,0,2,1,1}/16; mean 1.8, CI [0, 3.7]
+ ensemble disagreement, $\lambda \in \{0, \frac{1}{2}, 1, 2\}$	—	{2,2,2,1}/16
true-state cost (no pockets)	—	16/16

the counterfactual: nothing about the task, budget, or planner prevents success; only the pockets do.

Summary of the mechanism section. A frozen-base learned cost, under test-time optimization pressure at 100×6 candidates per replan, behaves like a reward model under over-optimization: its average-case accuracy is irrelevant because the optimizer samples from its worst case. Every repair that improves average-case metrics — readout precision, grounding, disagreement — leaves the worst case reachable. We were unable to produce a plannable post-hoc cost on a frozen action-conditioned JEPA encoder for contact manipulation, across every lever available to us; we conjecture, and our evidence supports, that **any post-hoc cost on such an encoder is planner-exploitable at contact**.

7 The axis that moves: counterfactual training of the predictor

The ladder bypassed the predictor (perfect dynamics); the deployed stack still uses one. Independently of the cost-side wall, the diagnostic of Sec. 3 identifies a training-target defect in the predictor: under a teacher-forced objective, the target latent already contains the action’s effect, so the objective admits solutions in which the predictor matches targets without using the action channel — a mechanism independently confirmed in a toy setting by (UWM-JEPA authors) (2026). The corrective is to make the objective *counterfactual over actions*: the prediction under the factual action must beat predictions under distractor actions, not just match its target.

We instantiate this as an InfoNCE-over-actions objective on the frozen DROID world model (`dino_wm_droid`): a predictor-LoRA of 0.469M parameters (encoder untouched, no object ground truth), trained so that $F(z_t, a_t)$ scores higher against z_{t+1} than $F(z_t, a_k^-)$ for executed-action distractors a_k^- , with a stop-gradient on the target. Counterfactual ranking accuracy rises $0.195 \rightarrow 0.736$ while reconstruction *improves* ($0.695 \times$ baseline error) — the objective is not trading prediction quality for action sensitivity; it is removing a shortcut.

The effect is robust to training seed, and transfers to a second model. Across four CF training seeds, the pooled improvement is tight and every seed beats the frozen baseline on every metric: Action Error $1.468 \rightarrow 1.076 \pm 0.012$, Action Score $0.466 \rightarrow 0.544 \pm 0.019$, effect-CRA $0.061 \rightarrow 0.610 \pm 0.034$ (mean $\pm$ sd over seeds; the single-seed column above sits at the seed mean). On a second predictor family (`jepa_wm_droid`), the objective lifts effect-CRA $0.076 \rightarrow 0.32$ and Action Score $0.476 \rightarrow 0.51$; Action Error also improves ($1.386 \rightarrow 1.27$) once the LoRA rank is raised to 16, confirming the mechanism is not specific to one checkpoint.

Two readings, stated carefully. Positively: this is the predictor-axis repair done at the correct level — the training objective, not a post-hoc patch — and it produces the only across-the-board improvement of any intervention in this paper, on real-robot data, without labels. Negatively: given Sec. 6, we do *not* claim this alone crosses the contact wall in closed-loop planning — the cost-side exploitation is untouched by a

Table 7: DROID planning probe, frozen vs. counterfactually-trained predictor (A/B, same protocol, $n=157$ planned transitions; CEM 64×15). Action Error = distance between the planned and the robot’s executed action, in normalized action units (lower is better); Action Score is the upstream aggregate (higher is better); effect-CRA as in Sec. 3.1 (chance ≈ 0.059). Action Error improves in all six regime $\times$ horizon cells (Action Score in 5/6); effect-CRA leaves the chance floor for the first time in this paper.

regime	H	Action Error $\downarrow$		Action Score $\uparrow$		effect-CRA $\uparrow$	
		frozen	CF	frozen	CF	frozen	CF
pre-grasp	1	1.044	0.792	0.621	0.653	0.050	0.538
gripper	1	1.080	0.577	0.607	0.748	0.000	0.583
contact	1	1.267	0.950	0.539	0.584	0.125	0.575
pre-grasp	3	1.573	1.166	0.428	0.490	0.050	0.662
gripper	3	2.208	1.571	0.197	0.313	0.045	0.727
contact	3	1.907	1.661	0.307	0.273	0.118	0.529
pooled		1.468	1.086	0.466	0.525	0.061	0.605

better predictor (the ladder proved planning fails identically under *perfect* prediction). The two axes are complementary and both necessary; conflating them is precisely the error the oracle ladder is designed to prevent.

8 Discussion, scope, and limitations

What follows for evaluation. Every quantity the field currently reports was satisfied at some rung of this paper while planning failed: held-out prediction error (Sec. 4 harness on-manifold), action grounding (94–100% in Table 6), readout precision (91.5% within 5 cm on the planner’s own query distribution). The quantity that tracked the failure throughout is the *exploitation gap*: the discrepancy between the cost’s claim and the world’s truth *on the states the optimizer converges to* (Table 5, and the overoptimization curves of Fig. 1). We propose reporting it directly: mine the planner’s elite population and decode it against ground truth (in sim), or against a held-out verifier (on real data). It is cheap — one extra decode pass over frames the planner already produced.

What follows for method design. The results sort the design space into three responses. (i) *Harden the cost*: our evidence says this loses on frozen-base representations — average metrics improve, worst-case pockets remain, the optimizer finds them (Table 6). (ii) *Fix the training objective*: the counterfactual objective (Sec. 7) is the right lever for the predictor axis, and encoder-level analogues remain open — but grounding alone demonstrably does not buy planning. (iii) *Remove the adversary*: if test-time search is what weaponizes cost error, amortizing planning into a policy or inverse-dynamics controller ((authors), 2026b), or anchoring the search to demonstration priors ((IMWM authors), 2026), removes the optimization pressure rather than the pockets. We tested this directly (Sec. 8): a goal-conditioned inverse-dynamics controller built from the same frozen representation and readout, run with *no* search, and the same controller with search re-added in increasing doses. The result is a null and it sharpens the paper’s claim rather than softening it — see below.

Removing the adversary is necessary, not sufficient. Under the latent oracle on push, the search-free amortized controller solves 0/8, and every re-added search dose (CEM iterations $\{0, 2, 6, 12, 24\}$ seeded from the controller) also solves 0/8; in all arms the object never leaves its start (med $\|obj - goal\| \approx 0.25$ m, Fig. 2A). Removing the optimizer does not cross the wall. Two things are nonetheless confirmed. On the mechanism side, the per-replan seed-vs-chosen comparison shows search “improves” the proxy over the seed on 98–100% of replans while *corrupting* the true state (proxy down, truth up) on a fraction that rises monotonically with dose ($0 \rightarrow 7\%$, Fig. 2B) — the adversary is real and dose-dependent even here. But it is not the *only* wall: a controller with no optimizer to exploit it still fails, because the frozen representation and its object readout do not carry a plan that moves the object. The honest synthesis across the oracle

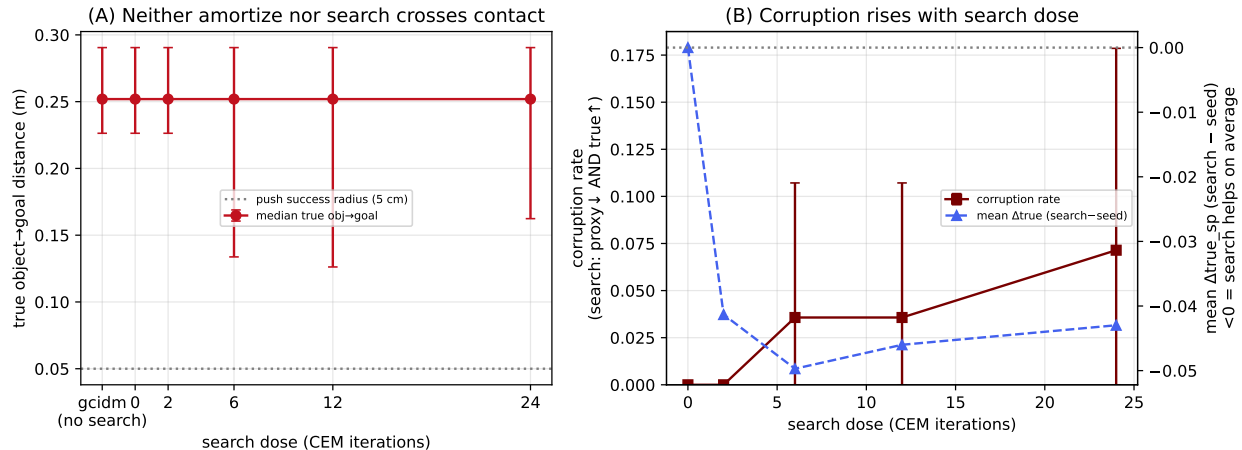


Figure 2: **Removing the search adversary does not cross contact (E1).** (A) True object→goal distance for the pure amortized controller (“gcidm”, no search) and for search re-added in doses; all sit at ≈ 0.25 m, far above the 5 cm success radius. (B) Per-replan corruption rate (search lowers the proxy *and* raises the true cost) rises with search dose (0 $\rightarrow$ 7%), while the mean true-cost change from search stays negative (on average search still helps the *hand*) — the adversary is real but small, and the object never moves regardless.

ladder, the mitigation table, and this experiment is that contact planning faces *two* walls — an exploitable cost/optimizer interface (E0, Secs. 5–6.2) and an insufficient frozen representation (E1) — and that *grounding is necessary but not sufficient*: neither hardening the cost, nor fine-tuning the encoder for grounding, nor removing the search adversary, crosses it alone.

Scope and honest limits. The exploitation mechanism is established on Metaworld, where simulator state lets us verify every claim against ground truth; on DROID we establish the diagnostic (scale-flat chance-floor grounding) and the predictor-axis repair, but closed-loop exploitation curves on real robots remain future work. The mitigation table covers the levers available on frozen pretrained checkpoints (readouts, metrics, adapters, LoRA fine-tuning, ensembles); full encoder retraining with planner-aware objectives is beyond our compute and remains open — our claim is about the *post-hoc* regime. The boundary-regime labels use an object-displacement proxy (no MuJoCo contact ground truth in the released data); mw-door-close is excluded as a proxy anomaly. The Reach parity check compares one released checkpoint over 16 episodes against a published 3-seed $\times$ 96-episode number — we read it as “reproduces, inside the CI”, not a seed-matched replication. Finally, an information-sufficiency question remains open beneath everything: vision-only latents may lack contact state outright ((authors), 2026a); our state-oracle row shows the *task* is solvable from privileged state, but whether any vision-only cost can be made pocket-free is not settled by this paper.

9 Conclusion

Contact-rich planning with JEPA world models does not fail for want of encoder scale, prediction accuracy, grounding, or readout precision — all were satisfied, measured, at some rung of our ladder while push stayed at 0–2/16. It fails because a test-time optimizer is pitted, hundreds of candidates per control step, against a learned cost whose residual-error pockets it is structurally guaranteed to find — and the same planner solves the same task 16/16 the moment the cost has no pockets. Repairs aimed at the average case (hardened readouts, fine-tuned encoders, ensembles) leave the worst case reachable; the training-side objective that repairs the predictor’s action channel is real and label-free but orthogonal to the cost-side wall. World models intended for planning should be evaluated the way their planners attack them: not by how well they predict, but by how little they can be exploited.

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A Boundary Blindness: definition

The instrument compares, on a neighbourhood of similar states, how much the *world’s* outcome fans out under nearby actions against how much the *model’s* prediction fans out under the same actions.

Neighbourhood. For each anchor transition i with latent z_i and action a_i , $\mathcal{N}(i)$ is its $M=16$ nearest cached transitions in latent distance, radius-limited with relaxation (the same neighbourhood the hard-negative CRA sampler uses). o_j is the true scalar outcome of neighbour j : object displacement $\|\Delta\text{obj}_j\|$ on Metaworld, the $\|\Delta z_j\|$ proxy on DROID.

Boundary regime. A transition is *boundary* when small action changes produce large true-outcome changes across its neighbourhood:

$$\text{boundary_score}_i = \frac{\text{std}_j o_j}{\text{mean}_j \|a_j - a_i\| + \varepsilon}, \quad (3)$$

thresholded at a population percentile. Free-space drift has large $\|\Delta z\|$ but smooth neighbourhoods and scores low.

The two sensitivities and BB. The world’s local action-sensitivity S_i^{true} is the dispersion of true outcomes across the neighbourhood; the model’s, S_i^{model} , applies each neighbour’s action to the *single* anchor state and measures the RMS spread of $F(z_i, a_j)$ around its centroid. Each is z-scored over the population and

$$\text{BB}_i = \text{relu}(\tilde{S}_i^{\text{true}} - \tilde{S}_i^{\text{model}}), \quad (4)$$

pooled (n_b -weighted) over the boundary mask with trajectory-clustered bootstrap CIs. Because BB references the world, a high value cannot be an artefact of small one-step latent deltas.

B The grounded-channel fix ladder (metric-level repair)

Deferred from Sec. 4: the model-side repair ladder that motivated the oracle ladder. **Nulls:** mixture-density predictor heads (soft-NLL/WTa/hard-EM/supervised, $K \in \{2, 3\}$) beat their $K=1$ NLL controls but do not move $\text{BB}_{\text{boundary}}$ — the bifurcation spans 9.9 L_2 units against a 106 median residual (the geometry under-weights the boundary subspace $10\times$), and the boundary’s action-dependence is unlearnable from expert-only data at the head level. Probe-subspace metric re-weighting redistributes BB without reducing it. **The probe chain:** object state is decodable from the frozen latent (V1 err 0.064 vs. sd 0.094), propagates through the predictor for *factual* actions (V2 0.059), but the *counterfactual* action→object response is noise (V3 spread-corr +0.035). **The repair:** a 0.5M grounded dynamics channel $h(z_t, a) \rightarrow \widehat{\Delta\text{obj}}$ (mean/max-pooled tokens + action MLP), trained on cached latents with everything frozen, restores counterfactual tracking (0.035→0.682 DINO-WM; 0.702 JEPA-WM), halves boundary BB on both families (1.323→0.660; 1.280→0.620), and improves the closed-loop cost surface (Table 3) — while flipping zero successes, the observation Sec. 5 explains. Honest negative: the recipe does not transfer to DROID’s noisy whole-state proxy label at 2.1k transitions (rank-corr +0.708 but spread magnitude collapsed; BB unchanged) — the principle needs a learnable grounding label.

C Reproduction integrity

Closed-loop sim evaluation is fragile, and getting the baseline wrong would invalidate the parity argument. Three environment-side bugs, found and fixed, all model-agnostic: **(1)** Metaworld’s offscreen renderer is constructed in the env constructor, so camera/size assignments afterwards were silently ignored (default camera at 480 px vs. the trained **corner2** at 224 px; symptom: one-step prediction error $8.5\times$ dataset level). We re-create the renderer explicitly, mirroring the upstream wrapper. **(2)** The released training frames are vertically flipped relative to today’s **corner2** render; pixel-calibration against dataset initial frames scores the flipped wrapper-camera render at MSE 71.6 vs. ≥ 3000 for every unflipped candidate. **(3)** The success flag fires at a 5 cm radius but the scripted expert refines to ~ 1 cm; taking the goal frame at first-success rather than the expert’s final frame parked the planner just outside the radius. **Verification:** after the fixes, one-step prediction error is $1.4\times$ dataset level, latent NN ratio 0.97, and replaying dataset actions reproduces the proprioceptive trajectory to 1.5 mm median end-effector error — only the visuals had drifted, never the physics. The reach baseline then recovers from near-zero to 37.5% (Table 3), inside the published CI.

D AUG and ECS: the raw-gap corroboration

CRA leads the main text because, as a ranking, it is cross-model comparable. The two raw-gap metrics corroborate it: AUG (prediction-error gap between permuted and factual actions; ≈ 0 iff actions ignored) and ECS (AUG on effect-bearing transitions), both on the hard nearest-neighbour family.

Table 8: AUG / ECS, hard nearest-neighbour family, per regime (n_{eff} -weighted pools; dash = no effect-bearing transitions). Raw latent gaps are *not* comparable across models (latent scales differ); the cross-model statement lives in CRA (Tables 1, 2).

model	free space	pre-grasp	gripper	contact
<i>DROID</i> (real robot; AUG / ECS)				
DINO-WM (22M)	0.038/–	0.011/0.0105	0.038/0.0645	0.070/0.0094
JEPA-WM (300M)	0.000/–	0.000/0.0002	0.001/0.0004	0.001/–0.0011
V-JEPA-2-AC (1B)	0.006/–	0.002/0.0016	0.003/0.0026	0.003/–0.0059
V-JEPA-2-AC-OSS (1B)	0.004/–	0.001/0.0010	0.001/0.0005	0.003/–0.0030
<i>Metaworld</i> (sim; AUG / ECS; pooled over 12 tasks)				
DINO-WM (22M)	0.088/0.1007	0.051/0.0997	–/–	0.081/0.0893
JEPA-WM (300M)	0.089/0.1020	0.058/0.1114	–/–	0.095/0.1052

The effect-conditioned action gap is statistically indistinguishable from zero on the real-robot data across the full $45\times$ scale range — the chance-level CRA reflects genuine action-ignoring, not a ranking quirk. Source: `results/{droid,metaworld}.diagnostic.csv`, `hard_nn` rows.

E Full per-task tables

[TODO: per-task BB CSV dumps; mitigation-arm episode tables (phaseC/D/F/G logs); E0 curve tables when jobs 24740/24741 land; Phase-H seed-sweep CIs.]